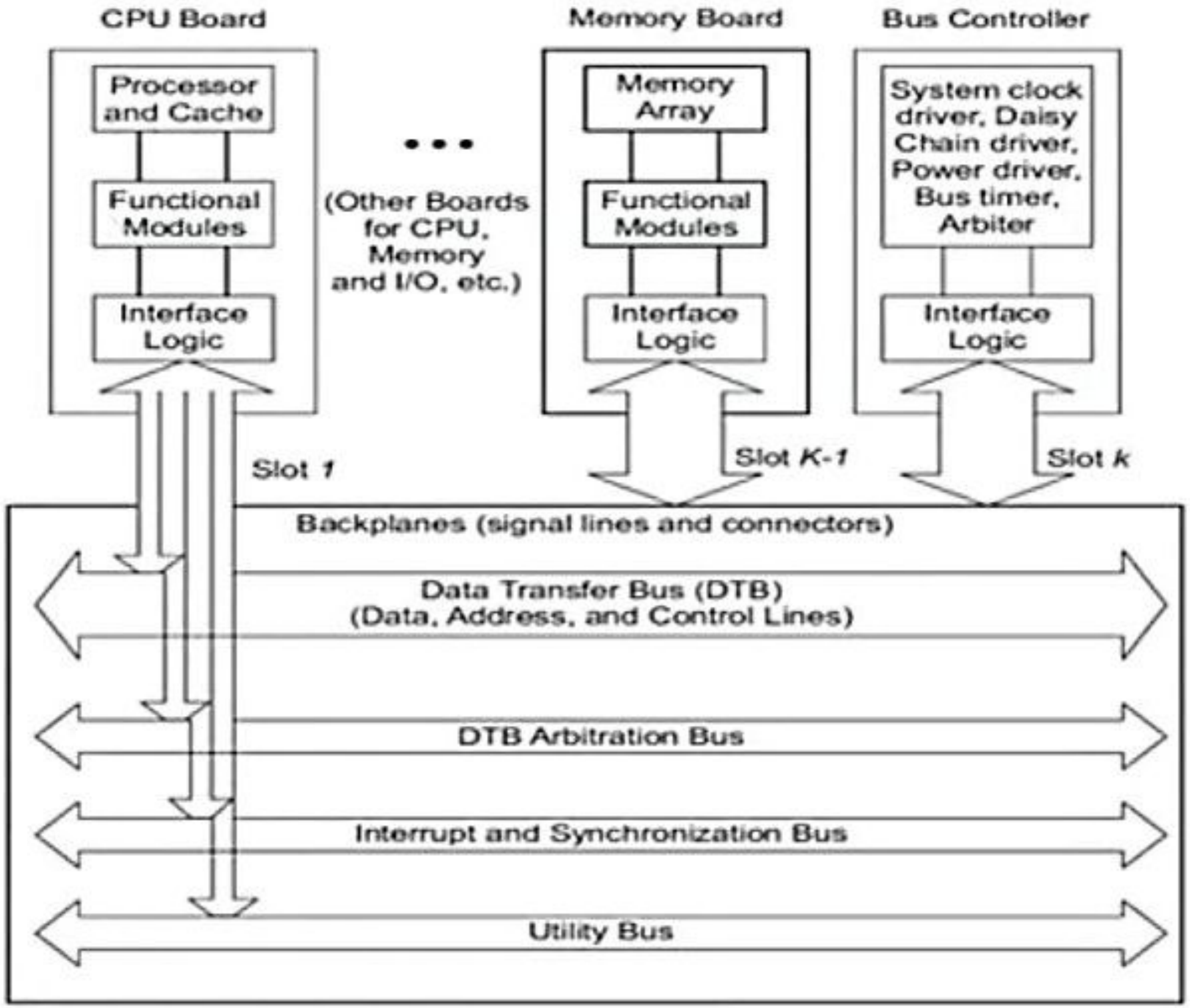
Bus Architecture

- The main function of Bus system is to interconnect main components of computer.
- The effective bandwidth available for each processor is inversely proportional to the number of processors contending for the bus. This is why most of the bus based commercial multiprocessors are small in size using 4 to 16 processors.
- At a time only one processor will be allocated with the access permission of bus.

* as more processors (CPUs) compete for shared bus access, the total available bandwidth gets divided, reducing the effective speed for each, which limits system size to about 4-16 processors before performance degrades significantly, this is also known as **bus contention.**

Backplane Bus



A backplane bus is a central communication backbone, essentially a large circuit board with parallel connectors, that links multiple functional modules (like CPUs, memory, I/O cards) in a system, allowing them to share data, power, and resources efficiently, creating a modular, scalable, and easily upgradeable hardware architecture

Data Transfer Bus:

- Data transfer bus includes Data, address and control lines in a VME (VERSA module Eurocard IEEE 1014) bus.
- The address bus is used to transfer data and device address.
- The data lines are proportional to the memory word length.

For eg. The VME bus specification has 32 address lines and 32 or 64 data lines so the 32 address lines can be multiplexed during data transfer cycles.

- Control lines send the control signal for enabling/disabling address and data lines.

Bus Arbitration Control:

The process of assigning control of Data transfer Bus (DTB) to a sequencer is called arbitration.

Interrupt lines: are used to handle interrupts, for that dedicated lines may be used to synchronize parallel activities among the processor module.

Utility Lines: include signals that provide periodic timing (clock)

Functional Modules:

- A functional module is the combination of electronic circuits that present in functional board.
- Some of the special functional modules are:
- **Arbiter:** An arbiter is a functional module that accepts bus requests from the requester module and grants control of the data transfer bus to one of the requester but one at a time.
- **Bus Timer:** it measures the time each data transfer takes on the data transfer bus (DTB) also it terminates the DTB cycle if transfer takes too long.
- **Interrupter:** this module generates an interrupt request and provides status information when an interrupt handler module requests it.
- **Location monitor:** is a functional module that monitors the data transfer over the DTB.
- **Power monitor:** it checks the status of the power source and signals when power becomes unstable.
- **System clock Driver:** this module provides a clock timing signal on the utility bus.

Types of backplanes

- **Active backplanes**:
- Contain components like processors, switches, and memory.
- Have electronic infrastructure to manage communication between slots.
- Have a non-zero risk of malfunction due to their complexity
- Can be used for systems that have many points of failure.
- **Passive backplanes**:
- Contain bus connectors, but no active components
- Have expansion boards that handle data communication
- Can be used for systems with a single point of failure

Other Bus Types in Backplane Architecture

- **Industry standard architecture (ISA):** An I/O device that can handle 16-bit data transfers
- **Extended ISA (EISA):** An enhanced version of the ISA bus that can handle 32-bit data transfers
- **Peripheral component interconnect (PCI):** A local bus system for high-end computers that can transfer 32 or 64 bits of data
- **Compact PCI (cPCI):** Uses the electrical standards of the PCI bus, but is packaged in a Versa Module Eurocard (VME) bus

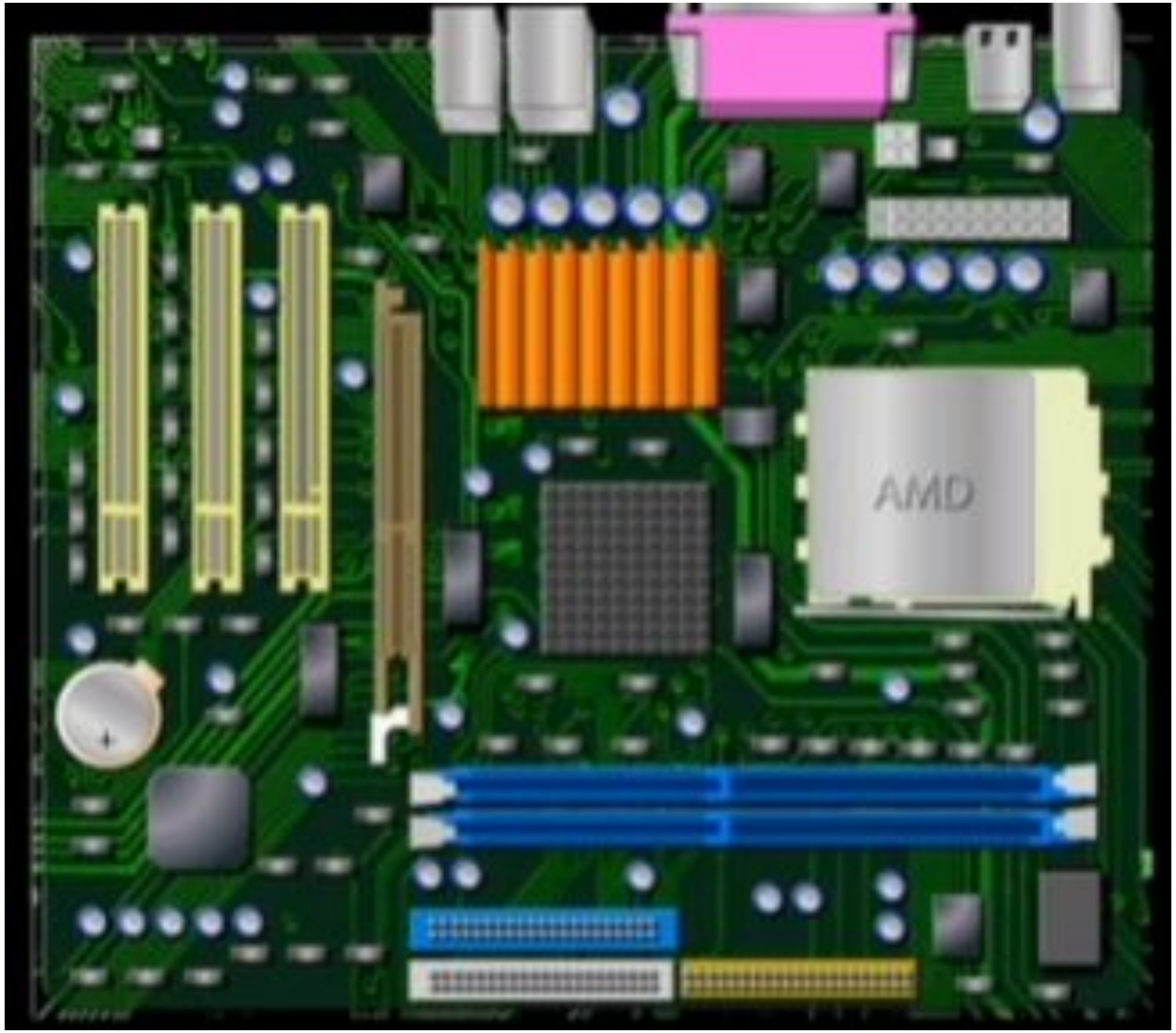
Limitations of Backplane Bus

- Due to electrical, mechanical and packaging limitations only a limited number of boards can be plugged into a single backplane.
- The bus system is difficult to scale mainly limited by contention and packaging constraints.

Key limitations

Bandwidth limitations:

- As all devices on the bus share the same communication channel, the overall bandwidth is divided among them, leading to potential bottleneck when many devices are actively transferring data simultaneously.
- **Signal degradation:**
- With more devices connected to the bus, signal integrity can deteriorate over long traces, leading to potential data corruption or transmission errors, especially at high speeds.
- **Scalability issues:**
- Expanding a backplane system by adding new devices can become challenging as the bus becomes increasingly congested, potentially requiring a complete system redesign to accommodate increased traffic.
- **Single point of failure:**
- If the backplane itself fails, all connected devices are affected, making it a critical component for system reliability.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